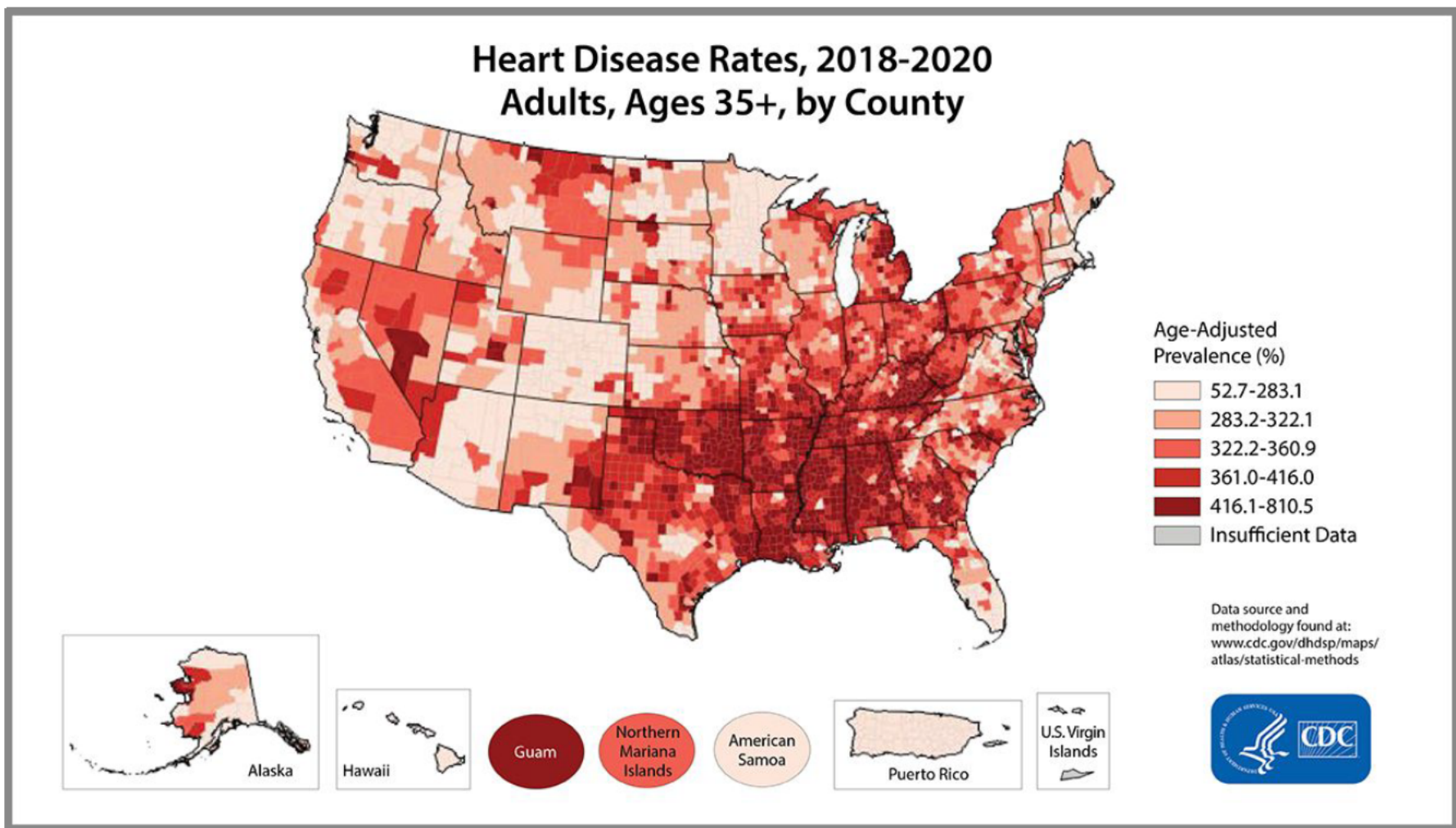


Using Heart Disease Status Using MLP and CF

Purpose



This project investigates the effectiveness of Collaborative Filtering and Multilayer Perceptrons in predicting heart disease status. We aim to assess whether these underexplored approaches can yield accurate and efficient results compared to conventional ML models used in past research.

Background

Classification model	Accuracy	Sensitivity	Specificity	AUC	Precision	F1-score	MCC
KNN (K = 7)	85.65	83.84	87.57	92.10	86.82	0.85	0.71
DT	84.28	83.84	84.76	88.68	84.26	0.84	0.68
ET	89.36	88.21	90.58	92.05	88.90	0.88	0.77
GB	88.47	89.54	87.37	92.69	86.39	0.88	0.77
RF	88.18	89.92	85.97	92.52	86.17	0.87	0.76
SVM (kernel = 'rbf')	80.57	83.26	77.75	88.03	80.06	0.81	0.61
AB	85.94	86.50	85.37	90.72	84.33	0.86	0.72
NB	82.62	83.84	81.36	86.81	82.76	0.83	0.65
LR (C = 10)	80.77	83.46	77.95	85.00	80.35	0.81	0.61
ANN (13, 20, 2)	87.59	88.02	86.57	92.40	87.58	0.87	0.74

Figure 1: Prior Study Classification Report (Muhammad et al. 2020)

Multilayer perceptron and collaborative filtering are not included in prior computational heart disease prediction studies. Our research seeks to add onto prior research to see if we could **increase efficiency** and **accuracy** of heart disearch prediction with MLP or CF.

Methods

CF

- **dataset:** heart disease prediction dataset with 21 columns, and 10,000 rows (users)
- **preprocessing:** one-hot encoding of categorical features, min-max scaling, random removal of 30% of dataset
- **item-item collaborative filtering:** item-item similarity matrix, min-max scale, compute weighted avg. for k-most similar items, round outcomes to 0 or 1 (target=heart disease status)
- **model optimization:** grid search for hyperparameter tuning of k (2-13) and similarity score method (cosine or L2)

MLP

- **dataset:** heart disease prediction dataset with 21 columns, and 10,000 rows (users)
- **preprocessing:** one-hot encoding of categorical features, mean imputation for missing values, standardization
- **multilayer perceptron:** 1 hidden layer, 10 nodes with ReLU activation, sigmoid output, binary cross-entropy loss
- **model optimization:** 500 epochs, learning rate = 0.01, He initialization
- **performance metrics:** accuracy, precision, recall, F1 score, log loss, ROC AUC

Results

CF

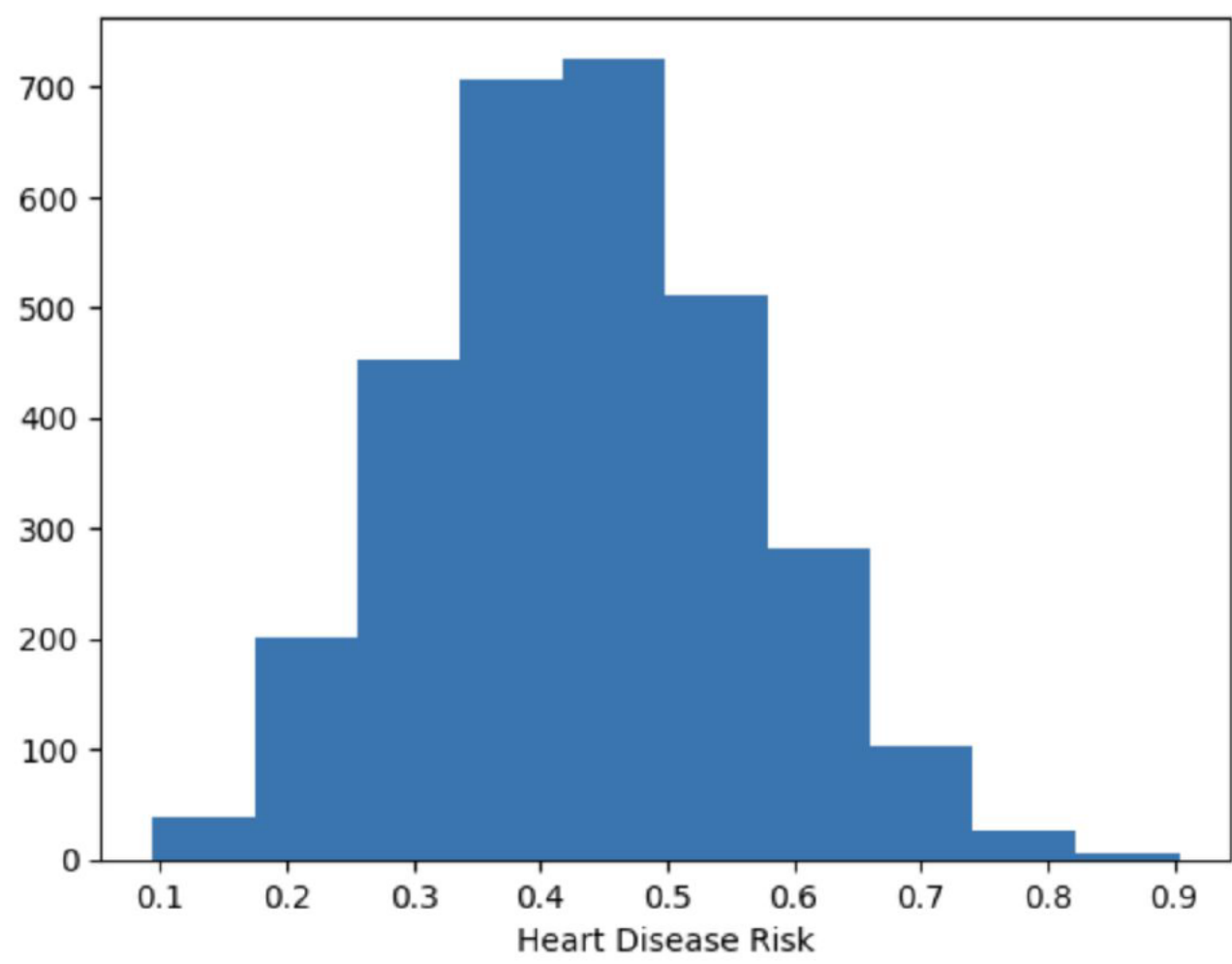


Figure 2: Distribution of Predicted Heart Disease Risk

	precision	recall	f1-score	support
0.0	0.81	0.7	0.75	2451
1.0	0.21	0.31	0.25	602
accuracy			0.63	3053

Figure 3: CF Classification Report

- **optimized model parameters:** item-item, cosine similarity, k=11, log loss = 0.655
- **top 5 similar features:** BMI (0.04), Female (0.04), Triglyceride level (0.04), High LDL Cholesterol (0.03), High Alcohol Consumption (0.03)
- **confusion matrix:** 1724 TN, 727 FN, 188 TP, 414 FP

MLP

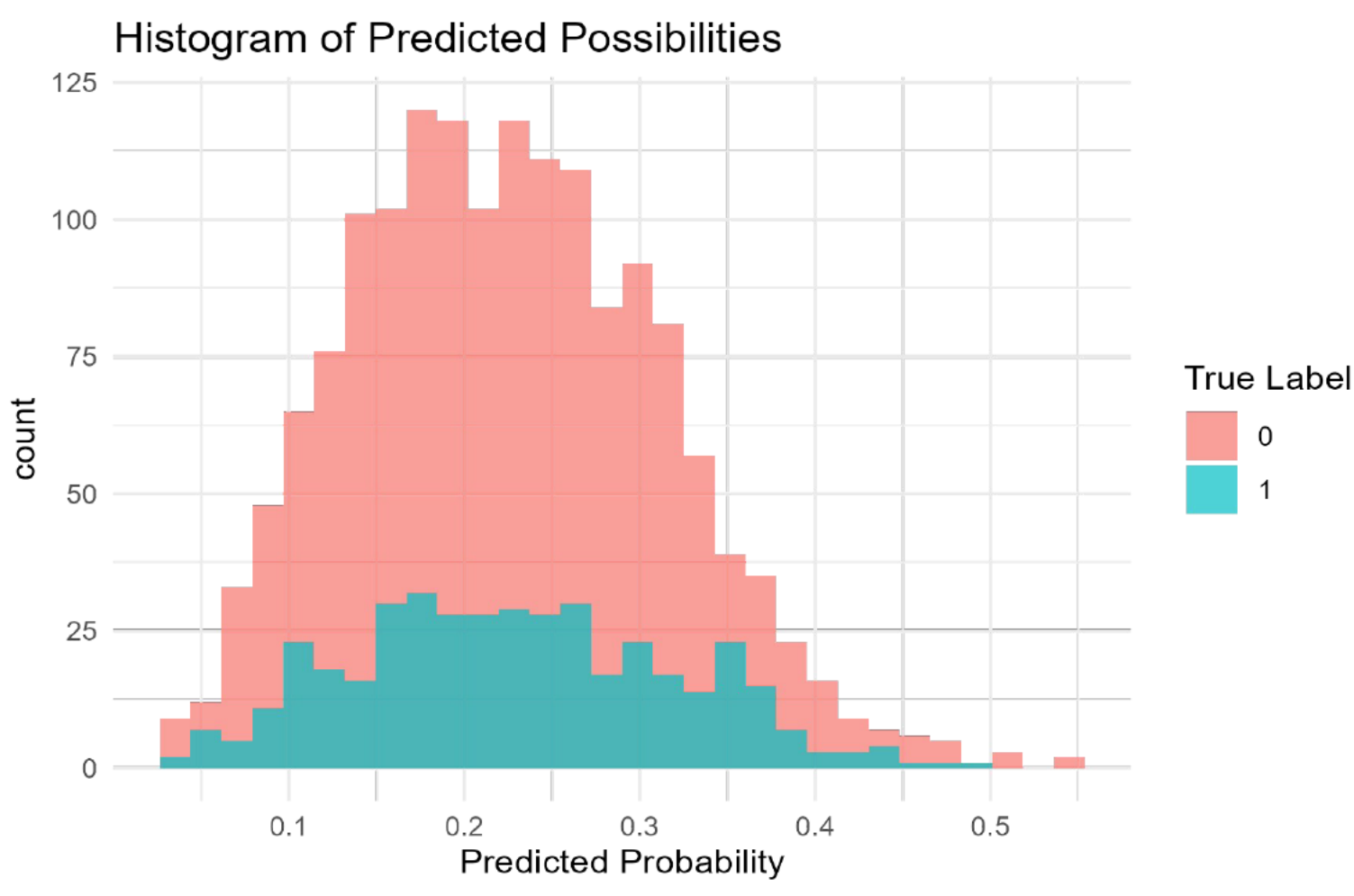


Figure 4: Histogram of Predicted Probabilities (Colored by True Label)

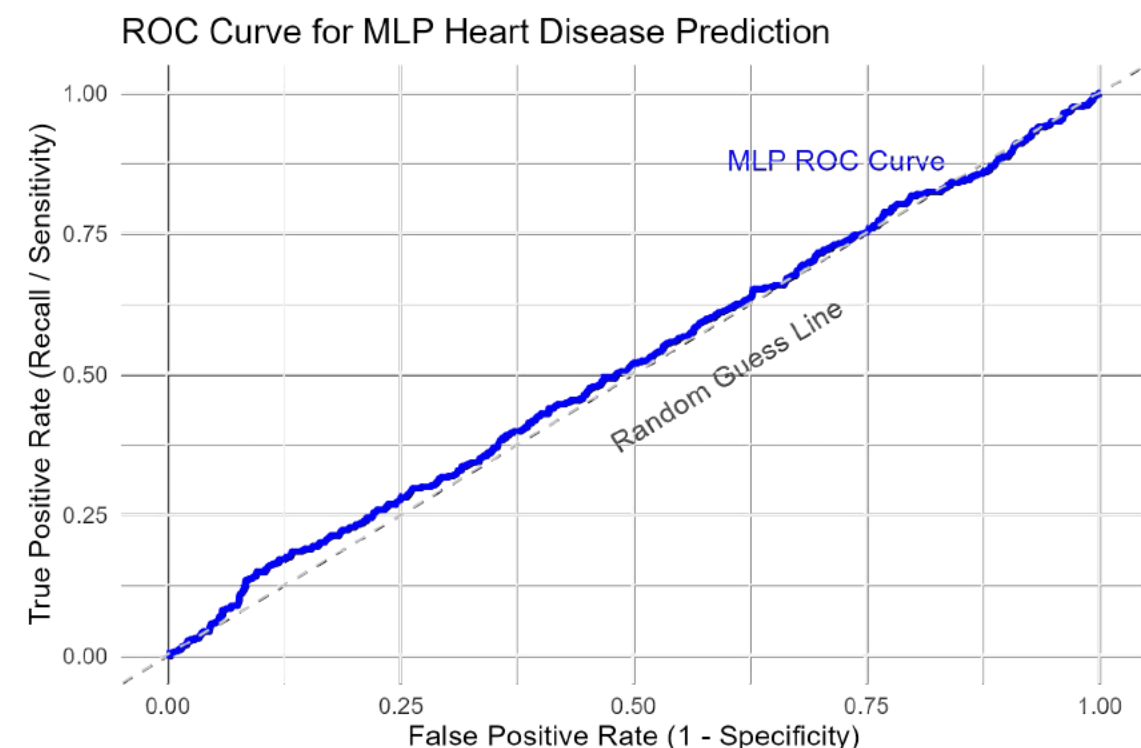


Figure 5: ROC Curve with Baseline

- **MLP metrics:** Accuracy = 0.79, Recall = 1.00 (class 0), 0.00 (class 1), F1 = 0.88, Log Loss = 0.5327, AUC = 0.5152
- Model struggled with positive class due to imbalance
 - High confidence predicting absence (Fig. 4)
 - ROC Curve close to baseline (Fig. 5)

Conclusions

- The MLP outperformed the CF model in both accuracy (0.79 vs. 0.63) and log loss (0.5327 vs. 0.655)
- Despite stronger performance, the MLP failed to detect any true positive cases
- Both models showed a strong bias toward the negative class, driven by class imbalance in the dataset
- Evaluation metrics beyond accuracy are essential for assessing model performance in imbalance settings

Next Steps

- **data remediation**
 - fix data imbalance (most individuals didn't have heart disease)
 - diversify dataset
- **model expansion**
 - User-user CF
 - MLP hyperparameter tuning
 - other ML models not covered in prior research
- **long term**
 - integrate with time series data
 - deploy model in a web app

References

Muhammad, Yar, et al. "Early and accurate detection and diagnosis of heart disease using intelligent computational model." *Scientific Reports*, vol. 10, no. 1, 12 Nov. 2020, <https://doi.org/10.1038/s41598-020-76635-9>.

Centers for Disease Control and Prevention. "Heart Disease Facts." Centers for Disease Control and Prevention, 7 May 2024, <https://www.cdc.gov/heart-disease/data-research/facts-stats/index.html>.